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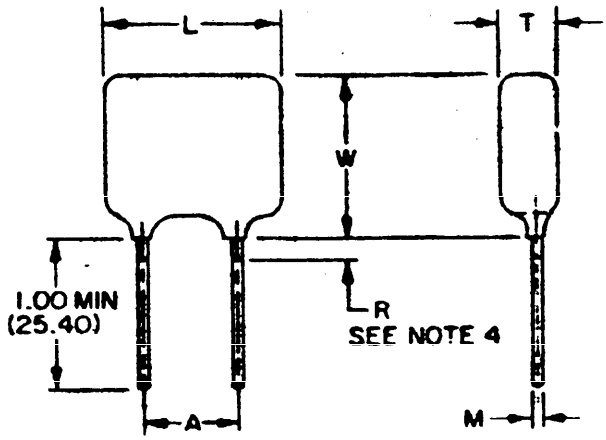
MILITARY SPECIFICATION SHEET

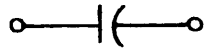
CAPACITORS, FIXED, MICA DIELECTRIC, ESTABLISHED RELIABILITY,

STYLES CMR03, CMR04, CMR05, CMR06, CMR07, AND CMR08

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39001.





CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches (see table I).
2. Millimeters are in parentheses (see table I).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. The 'R' dimension may not be solderable as it may be covered by a clear epoxy or resinous coating.

FIGURE 1. Styles CMR03, CMR04, CMR05, CMR06, CMR07, and CMR08 capacitors.

FSC 5910

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TABLE I. Dimensions

STYLE CMR05												
Capacitance (pF)	300 V			100 V			50 V			300 V, 100 V, and 50 V		
	L	W	T	L	W	T	L	W	T	R MAX	A spacing ±.031 (.79)	M DIA ±.002 (.05)
	MAXIMUM			MAXIMUM			MAXIMUM					
1 to 12 incl	.270 (6.86)	.190 (4.83)	.110 (2.79)							.078 (1.98)	.120 (3.05)	.016 (.36)
15	.270 (6.86)	.190 (4.83)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)						
18 & 20	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)						
22 & 24	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)			
27	.270 (6.86)	.200 (5.08)	.130 (3.30)	.270 (6.86)	.190 (4.83)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)			
30 & 33	.270 (6.86)	.200 (5.08)	.130 (3.30)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)			
36	.270 (6.86)	.210 (5.33)	.130 (3.30)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.110 (2.79)			
39	.270 (6.86)	.210 (5.33)	.130 (3.30)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.120 (3.05)			
43	.270 (6.86)	.210 (5.33)	.140 (3.56)	.270 (6.86)	.200 (5.08)	.120 (3.05)	.270 (6.86)	.190 (4.83)	.120 (3.05)			
47 & 51	.270 (6.86)	.210 (5.33)	.140 (3.56)	.270 (6.86)	.200 (5.08)	.130 (3.30)	.270 (6.86)	.190 (4.83)	.120 (3.05)			
56	.270 (6.86)	.220 (5.59)	.150 (3.81)	.270 (6.86)	.200 (5.08)	.130 (3.30)	.270 (6.86)	.190 (4.83)	.120 (3.05)			
62	.270 (6.86)	.220 (5.59)	.150 (3.81)	.270 (6.86)	.210 (5.33)	.130 (3.30)	.270 (6.86)	.200 (5.08)	.120 (3.05)			
68	.270 (6.86)	.220 (5.59)	.150 (3.81)	.270 (6.86)	.210 (5.33)	.140 (3.56)	.270 (6.86)	.200 (5.08)	.120 (3.05)			
75 & 82	.270 (6.86)	.230 (5.84)	.160 (4.06)	.270 (6.86)	.210 (5.33)	.140 (3.56)	.270 (6.86)	.200 (5.08)	.120 (3.05)			
91	.270 (6.86)	.230 (5.84)	.170 (4.32)	.270 (6.86)	.210 (5.33)	.140 (3.56)	.270 (6.86)	.200 (5.08)	.130 (3.30)			
100 & 110	.270 (6.86)	.240 (6.10)	.180 (4.57)	.270 (6.86)	.220 (5.59)	.150 (3.56)	.270 (6.86)	.200 (5.08)	.130 (3.30)			
120	.270 (6.86)	.250 (6.35)	.190 (4.83)	.270 (6.86)	.220 (5.59)	.160 (4.06)	.270 (6.86)	.200 (5.08)	.130 (3.30)			
130				.270 (6.86)	.230 (5.84)	.160 (4.06)	.270 (6.86)	.210 (5.33)	.130 (3.30)			
150				.270 (6.86)	.230 (5.84)	.170 (4.32)	.270 (6.86)	.210 (5.33)	.140 (3.56)			
160				.270 (6.86)	.240 (6.10)	.170 (4.32)	.270 (6.86)	.210 (5.33)	.140 (3.56)			
170 & 180				.270 (6.86)	.240 (6.10)	.180 (4.57)	.270 (6.86)	.210 (5.33)	.140 (3.56)			
200				.270 (6.86)	.250 (6.35)	.190 (4.83)	.270 (6.86)	.220 (5.59)	.150 (3.81)			
220							.270 (6.86)	.220 (5.59)	.150 (3.81)			
240							.270 (6.86)	.220 (5.59)	.160 (4.06)			
270							.270 (6.86)	.230 (5.84)	.160 (4.06)			
300							.270 (6.86)	.230 (5.84)	.170 (4.32)			
330 & 360							.270 (6.86)	.240 (6.10)	.180 (4.57)			
390 & 400							.270 (6.86)	.250 (6.35)	.190 (4.83)	.078 (1.98)	.120 (3.05)	.016 (.36)

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TABLE I. Dimensions - Continued.

Capacitance (pF)	L		T		B, max	R, max	A Spacing ±.031(.79)	W, Dia ±.002(.05)
	Min	Max	Min	Max				
STYLE CMR07								
1 to 24, incl	.300 (7.62)	.360 (9.14)	.100(2.54)	.190 (4.83)	.330 (8.38)	.125(3.18)	.150 (3.81)	.016 (.36)
27	.300 (7.62)	.370 (9.40)	.100(2.54)	.190 (4.83)	.330 (8.38)	.125(3.18)	.150 (3.81)	.016 (.36)
30 to 68, incl	.300 (7.62)	.370 (9.40)	.100(2.54)	.190 (4.83)	.340 (8.64)	.125(3.18)	.150 (3.81)	.016 (.36)
75	.300 (7.62)	.370 (9.40)	.110(2.79)	.200 (5.08)	.340 (8.64)	.125(3.18)	.150 (3.81)	.016 (.36)
82	.300 (7.62)	.370 (9.40)	.110(2.79)	.200 (5.08)	.350 (8.89)	.125(3.18)	.150 (3.81)	.016 (.36)
91 & 100,	.300 (7.62)	.370 (9.40)	.110(2.79)	.200 (5.08)	.350 (8.89)	.125(3.18)	.150 (3.81)	.016 (.36)
110 & 120,	.300 (7.62)	.380 (9.65)	.110(2.79)	.200 (5.08)	.350 (8.89)	.125(3.18)	.150 (3.81)	.016 (.36)
130	.300 (7.62)	.380 (9.65)	.110(2.79)	.200 (5.08)	.360 (9.14)	.125(3.18)	.150 (3.81)	.016 (.36)
150 & 160,	.300 (7.62)	.380 (9.65)	.120(3.05)	.210 (5.33)	.360 (9.14)	.125(3.18)	.150 (3.81)	.016 (.36)
180	.300 (7.62)	.390 (9.91)	.120(3.05)	.210 (5.33)	.370 (9.40)	.125(3.18)	.150 (3.81)	.016 (.36)
200 to 390, incl	.300 (7.62)	.390 (9.91)	.140(3.56)	.220 (5.59)	.380 (9.65)	.125(3.18)	.150 (3.81)	.016 (.36)
STYLE CMR08								
1 to 62, incl	.400(10.16)	.450(11.43)	.090(2.29)	.170 (4.32)	.360 (9.14)	.125(3.18)	.225 (5.72)	.025 (.64)
68 to 82, incl	.400(10.16)	.450(11.43)	.095(2.30)	.180 (4.57)	.360 (9.14)	.125(3.18)	.225 (5.72)	.025 (.64)
91 & 100,	.400(10.16)	.460(11.68)	.095(2.30)	.180 (4.57)	.360 (9.14)	.125(3.18)	.225 (5.72)	.025 (.64)
110 to 130, incl	.400(10.16)	.460(11.68)	.095(2.30)	.180 (4.57)	.370 (9.40)	.125(3.18)	.225 (5.72)	.025 (.64)
150 to 180, incl	.400(10.16)	.460(11.68)	.100(2.79)	.190 (4.83)	.370 (9.40)	.125(3.18)	.225 (5.72)	.025 (.64)
200	.400(10.16)	.460(11.68)	.110(2.79)	.190 (4.83)	.380 (9.65)	.125(3.18)	.225 (5.72)	.025 (.64)
220 & 240,	.400(10.16)	.460(11.68)	.110(2.79)	.200 (5.08)	.380 (9.65)	.125(3.18)	.225 (5.72)	.025 (.64)
270 to 330, incl	.400(10.16)	.470(11.94)	.110(2.79)	.210 (5.33)	.390 (9.91)	.125(3.18)	.225 (5.72)	.025 (.64)
360 & 390,	.400(10.16)	.470(11.94)	.120(3.05)	.220 (5.59)	.400(10.16)	.125(3.18)	.225 (5.72)	.025 (.64)
STYLE CMR06								
430 & 470,	.550(13.97)	.640(16.26)	.090(2.29)	.200 (5.08)	.510(12.95)	.141(3.58)	.350 (8.89)	.032 (.81)
510 to 620, incl	.560(14.22)	.650(16.51)	.090(2.29)	.200 (5.08)	.510(12.95)	.141(3.58)	.350 (8.89)	.032 (.81)
680 to 910, incl	.560(14.22)	.650(16.51)	.100(2.54)	.210 (5.33)	.510(12.95)	.141(3.58)	.350 (8.89)	.032 (.81)
1000 & 1100,	.560(14.22)	.650(16.51)	.110(2.79)	.220 (5.59)	.520(13.21)	.141(3.58)	.350 (8.89)	.032 (.81)
1200 & 1300,	.570(14.48)	.660(16.78)	.110(2.79)	.220 (5.59)	.520(13.21)	.141(3.58)	.350 (8.89)	.032 (.81)
1500	.570(14.48)	.660(16.78)	.120(3.05)	.230 (5.84)	.520(13.21)	.141(3.58)	.350 (8.89)	.032 (.81)
1600	.570(14.48)	.660(16.78)	.120(3.05)	.230 (5.84)	.530(13.46)	.141(3.58)	.350 (8.89)	.032 (.81)
1800 & 2000,	.580(14.73)	.670(17.02)	.130(3.30)	.240 (6.10)	.530(13.46)	.141(3.58)	.350 (8.89)	.032 (.81)
2200	.580(14.73)	.670(17.02)	.140(3.56)	.250 (6.35)	.530(13.46)	.141(3.58)	.350 (8.89)	.032 (.81)
2400	.580(14.73)	.670(17.02)	.150(3.81)	.260 (6.60)	.540(13.72)	.141(3.58)	.350 (8.89)	.032 (.81)
2700	.590(14.99)	.680(17.27)	.160(4.06)	.270 (6.86)	.540(13.72)	.141(3.58)	.350 (8.89)	.032 (.81)
3000	.590(14.99)	.680(17.27)	.170(4.32)	.280 (7.11)	.550(13.97)	.141(3.58)	.350 (8.89)	.032 (.81)
3300	.590(14.99)	.680(17.27)	.180(4.57)	.290 (7.37)	.550(13.97)	.141(3.58)	.350 (8.89)	.032 (.81)
3600	.590(14.99)	.680(17.27)	.190(4.83)	.300 (7.62)	.560(14.22)	.141(3.58)	.350 (8.89)	.032 (.81)
3900	.600(15.24)	.690(17.53)	.200(5.08)	.310 (7.87)	.560(14.22)	.141(3.58)	.350 (8.89)	.032 (.81)
4300	.600(15.24)	.690(17.53)	.220(5.59)	.330 (8.38)	.570(14.48)	.141(3.58)	.350 (8.89)	.032 (.81)
4700	.610(15.49)	.700(17.78)	.220(5.59)	.350 (8.89)	.580(14.73)	.141(3.58)	.350 (8.89)	.032 (.81)
STYLE CMR07								
5100	.680(17.27)	.780(19.81)	.150(3.81)	.280 (7.11)	.860(21.84)	.141(3.58)	.425(10.80)	.040(1.02)
5600 & 6200,	.680(17.27)	.780(19.81)	.150(3.81)	.290 (7.37)	.870(22.10)	.141(3.58)	.425(10.80)	.040(1.02)
6800	.680(17.27)	.780(19.81)	.150(3.81)	.300 (7.62)	.870(22.10)	.141(3.58)	.425(10.80)	.040(1.02)
7500	.690(17.53)	.790(20.07)	.150(3.81)	.310 (7.87)	.880(22.35)	.141(3.58)	.425(10.80)	.040(1.02)
8200	.690(17.53)	.790(20.07)	.150(3.81)	.320 (8.13)	.880(22.35)	.141(3.58)	.425(10.80)	.040(1.02)
9100	.690(17.53)	.790(20.07)	.150(3.81)	.330 (8.38)	.880(22.35)	.141(3.58)	.425(10.80)	.040(1.02)
10,000	.700(17.78)	.800(20.32)	.200(5.08)	.340 (8.64)	.890(22.61)	.141(3.58)	.425(10.80)	.040(1.02)
11,000	.700(17.78)	.800(20.32)	.200(5.08)	.350 (8.89)	.890(22.61)	.141(3.58)	.425(10.80)	.040(1.02)
12,000	.700(17.78)	.800(20.32)	.200(5.08)	.360 (9.14)	.890(22.61)	.141(3.58)	.425(10.80)	.040(1.02)
13,000	.710(18.03)	.810(20.57)	.200(5.08)	.370 (9.40)	.890(22.61)	.141(3.58)	.425(10.80)	.040(1.02)
15,000	.710(18.03)	.810(20.57)	.250(6.35)	.390 (9.91)	.900(22.86)	.141(3.58)	.425(10.80)	.040(1.02)
16,000	.720(18.29)	.820(20.83)	.250(6.35)	.410(10.41)	.900(22.86)	.141(3.58)	.425(10.80)	.040(1.02)
18,000	.720(18.29)	.820(20.83)	.250(6.35)	.430(10.92)	.910(23.11)	.141(3.58)	.425(10.80)	.040(1.02)
20,000	.730(18.54)	.830(21.08)	.250(6.35)	.450(11.43)	.920(23.37)	.141(3.58)	.425(10.80)	.040(1.02)
STYLE CMR08								
22,000	1.300(33.02)	1.420(36.07)	.180(4.57)	.310 (7.87)	.880(22.35)	.141(3.58)	1.050(26.67)	.040(1.02)
24,000	1.310(33.27)	1.430(36.32)	.180(4.57)	.320 (8.13)	.880(22.35)	.141(3.58)	1.050(26.67)	.040(1.02)
27,000	1.320(33.53)	1.430(36.32)	.180(4.57)	.330 (8.38)	.880(22.35)	.141(3.58)	1.050(26.67)	.040(1.02)
30,000	1.330(33.78)	1.440(36.58)	.180(4.57)	.350 (8.89)	.890(22.61)	.141(3.58)	1.050(26.67)	.040(1.02)
33,000	1.330(33.78)	1.440(36.58)	.180(4.57)	.360 (9.14)	.890(22.61)	.141(3.58)	1.050(26.67)	.040(1.02)
36,000	1.340(34.04)	1.450(36.83)	.180(4.57)	.380 (9.65)	.900(22.86)	.141(3.58)	1.050(26.67)	.040(1.02)
39,000	1.340(34.04)	1.450(36.83)	.220(5.59)	.400(10.16)	.900(22.86)	.141(3.58)	1.050(26.67)	.040(1.02)
43,000	1.350(34.29)	1.460(37.08)	.225(5.72)	.420(10.67)	.910(23.11)	.141(3.58)	1.050(26.67)	.040(1.02)
47,000	1.360(34.54)	1.470(37.34)	.250(6.35)	.450(11.43)	.910(23.11)	.141(3.58)	1.050(26.67)	.040(1.02)
51,000	1.370(34.80)	1.480(37.59)	.250(6.35)	.470(11.94)	.920(23.37)	.141(3.58)	1.050(26.67)	.040(1.02)
56,000	1.350(34.29)	1.460(37.08)	.225(5.72)	.420(10.67)	.910(23.11)	.141(3.58)	1.050(26.67)	.040(1.02)
62,000	1.360(34.54)	1.470(37.34)	.250(6.35)	.450(11.43)	.920(23.37)	.141(3.58)	1.050(26.67)	.040(1.02)
68,000	1.370(34.80)	1.480(37.59)	.260(6.60)	.470(11.94)	.920(23.37)	.141(3.58)	1.050(26.67)	.040(1.02)
75,000	1.360(34.54)	1.470(37.34)	.240(6.10)	.440(11.18)	.910(23.11)	.141(3.58)	1.050(26.67)	.040(1.02)
82,000	1.370(34.80)	1.480(37.59)	.310(7.87)	.460(11.68)	.920(23.37)	.141(3.58)	1.050(26.67)	.040(1.02)
91,000	1.390(35.31)	1.500(38.10)	.340(8.64)	.500(12.70)	.940(23.88)	.141(3.58)	1.050(26.67)	.040(1.02)

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REQUIREMENTS:

Dimensions: See figure 1 and table 1.

Case material: Resin (dipped).

Capacitance value: See table II.

Capacitance tolerance: See table II.

DC rated voltage: 50 V (Y); 100 V (A); 300 V (C); or 500 V (D), as shown in table II.

Operating temperature range: See table II.

Characteristic: See table II.

Failure rate level: See table II.

Dielectric withstanding voltage: Method 301 of MIL-STD-202: 200 percent of rated dc voltage for not less than 1 second nor more than 5 seconds.

Barometric pressure: Method 105, condition D (100,000 ft), of MIL-STD-202. Test potential - 100 percent of rated dc voltage for style CMR04, 500 volts; 150 percent for lesser voltages and all other styles.

Insulation resistance (IR): Method 302, test condition A, of MIL-STD-202: (Minimum requirements):
- At 25°C:
  - 0 to 10,000 pF - - - - - - - - - 100,000 megohms
  - 10,000 pF and greater- - - - - 1,000 megohm-microfarads
- At 125°C:
  - 0 to 3,300 pF- - - - - - - - - - 10,000 megohms
  - 3,300 pF and greater - - - - - 33 megohm-microfarads
- At 150°C:
  - 0 to 1,500 pF- - - - - - - - - - 5,000 megohms
  - 1,500 pF and greater - - - - - 7.5 megohm-microfarads

Capacitance (Cap.): Method 305 of MIL-STD-202.

Dissipation factor (DF): See MIL-C-39001.

High voltage stabilization: 200 percent of rated dc voltage for 48 hours at a minimum temperature of 125°C or 150°C, as applicable.

Solderability: Method 208 of MIL-STD-202.

Vibration, high frequency: Method 204 of MIL-STD-202.

Temperature coefficient and capacitance drift: Within the limits specified for the applicable characteristic.

Thermal shock and immersion:
- Thermal shock: Method 107, condition A, MIL-STD-202, except high temperature shall be as follows:
  - Symbol O - +125°C.
  - Symbol P - +150°C.
- Immersion: Method 104, condition B, of MIL-STD-202.
  - DWV - Meet initial requirement.
  - IR - At least 30 percent of initial requirement.
  - ΔCap. - Shall not exceed ±1 percent or 1 pF, whichever is greater, of reference measurement.
  - DF - Not greater than 150 percent of initial requirement.

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Shock, specified pulse: Method 213, condition I (100 g(pk), sawtooth), of MIL-STD-202.

Terminal strength:
Pull test - Except for style CMR03, 5 pounds pull applied perpendicular to the leads for at least 5 seconds.
For style CMR03, pull shall be applied in the direction of the leads.
Twist test - 5 alternate 360° rotations.

Moisture resistance: Method 106 of MIL-STD-202.
DWV - Meet initial requirement.
IR - At least 25 percent of initial requirement.
ΔCap. - Shall not exceed ±1 percent or 0.1 pF, whichever is greater, of reference measurement.
DF - Not greater than 150 percent of initial requirement.

Life:
Forty-eight hours, minimum, at -55° $^{+0°}_{-2°}$C immediately followed by qualification or quality conformance test, as applicable.
Qualification test: 2,000 hours at applicable high test temperature with 100 percent or 150 percent of rated dc voltage applied, as applicable.
DWV - Meet initial requirement.
IR - Meet initial requirement.
ΔCap. - Shall not exceed ±1 percent or 1 pF, whichever is greater, of the initial measurement.
DF - Not greater than 150 percent of initial requirement.
Quality conformance test (240- and 2000-hour): 240 hours (L FR level) and 2000 hours (M, P, R, and S FR levels) at applicable high test temperature and 150 percent of rated dc voltage; requirements as specified for qualification.
Extended life (L FR level only):
Qualification test: 48,000 $^{+96}_{-0}$ hours at applicable high test temperature and rated dc voltage; requirements as specified for qualification.
Quality conformance test: 1,760 hours at applicable high test temperature and 150 percent of rated dc voltage; requirements as specified for qualification.

Fungus: Method 508 of MIL-STD-810 or manufacturer's equivalent.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202.
IR - Meet initial requirement.
ΔCap. - Shall not exceed ±1 percent or 1 pF, whichever is greater, of the initial measurement.
DF - Meet initial requirement.

Marking: Capacitors shall be marked as indicated in examples of figure 2.

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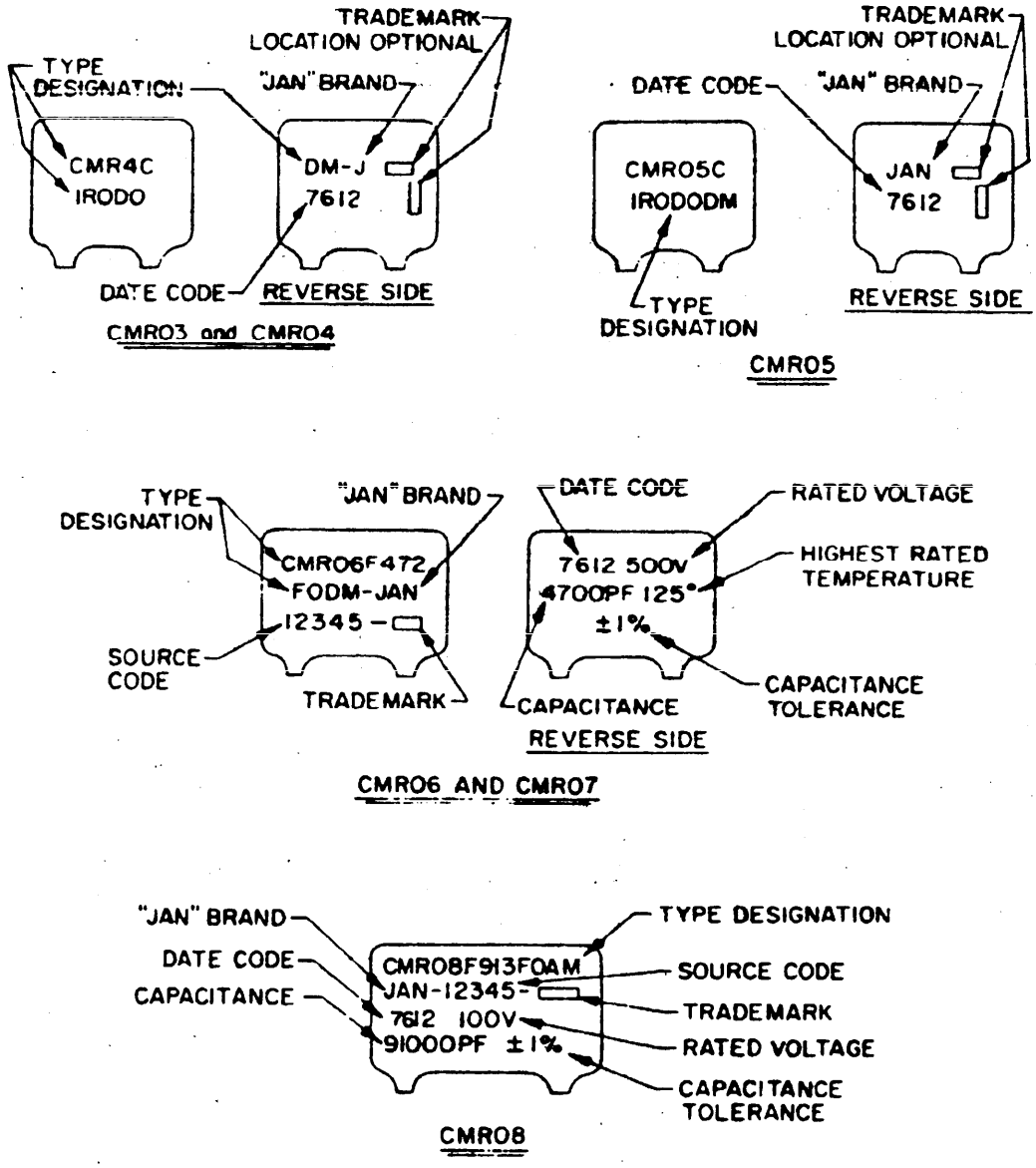


FIGURE 2. Marking examples.

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TABLE II. Type designations.

STYLE CMR03				
Type designation 1/	Capacitance value	DC rated voltage	Operating temperature range	Capacitance tolerance available
	(pF)	(volts)		
CMR03C1R0DOC-	1.0	300	0	D
CMR03C1R5DOC-	1.5	↓	↓	↓
CMR03C2R0DOC-	2.0			
CMR03C2R5DOC-	2.5			
CMR03C3R0DOC-	3.0			
CMR03C3R5DOC-	3.5			
CMR03C4R0DOC-	4.0			
CMR03C4R5DOC-	4.5			
CMR03C5R0DOC-	5.0			
CMR03C6R0DOC-	6.0			
CMR03C7R0DOC-	7.0			
CMR03C8R0DOC-	8.0			
CMR03C9R0DOC-	9.0			
CMR03C100DOC-	10			
CMR03C110DOC-	11	↓		
CMR03C120DOC-	12	300		
CMR03C150DO--	15	300, 100		
CMR03C180DO--	18	300, 100		
CMR03E200DO--	20	300, 100		
CMR03E220DO--	22	300, 100, 50		V
CMR03E240DO--	24	↓		D
CMR03E270-0--	27			GJ
CMR03E300-0--	30			
CMR03E330-0--	33			
CMR03E360-0--	36			
CMR03E390-0--	39			
CMR03E430-0--	43			
CMR03E470-0--	47			V
CMR03E510-0--	51			GJ
CMR03E560JO--	56			J
CMR03E620JO--	62			↓
CMR03E680JO--	68			
CMR03E750JO--	75			
CMR03E820JO--	82			
CMR03F910JO--	91			
CMR03F101JO--	100			
CMR03F111JO--	110	↓		
CMR03F121JO--	120	300, 100, 50		
CMR03F131JO--	130	100, 50		
CMR03F151JO--	150	↓		
CMR03F161JO--	160			
CMR03F171JO--	170			
CMR03F181JO--	180	↓		
CMR03F201JO--	200	100, 50		
CMR03F221JOY-	220	50		
CMR03F241JOY-	240	↓		
CMR03F271JOY-	270			
CMR03F301JOY-	300			
CMR03F331JOY-	330			
CMR03F361JOY-	360			
CMR03F391JOY-	390	↓	↓	↓
CMR03F401JOY-	400	50	0	J

1/ See footnote at end of table.

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TABLE II. Type designations - Continued.

STYLE CMR04				
Type designation 1/	Capacitance value	DC rated voltage	Operating temperature range	Capacitance tolerance available
	(pF)	(volts)		
CMR04C1R0D-D-	1.0	500	OP	D
CMR04C1R5D-D-	1.5	↓	↓	↓
CMR04C2R0D-D-	2.0			
CMR04C2R5D-D-	2.5			
CMR04C3R0D-D-	3.0			
CMR04C3R5D-D-	3.5			
CMR04C4R0D-D-	4.0			
CMR04C4R5D-D-	4.5			
CMR04C5R0D-D-	5.0			
CMR04C6R0D-D-	6.0			
CMR04C7R0D-D-	7.0			
CMR04C8R0D-D-	8.0			
CMR04C9R0D-D-	9.0			
CMR04C100D-D-	10			V
CMR04C110D-D-	11			D
CMR04C120J-D-	12			J
CMR04C150J-D-	15			↓
CMR04C180J-D-	18			
CMR04E200J-D-	20			
CMR04E220J-D-	22			V
CMR04E240J-D-	24			J
CMR04E270--D-	27			GJ
CMR04E300--D-	30			↓
CMR04E330--D-	33			
CMR04E360--D-	36			
CMR04E390--D-	39			
CMR04E430--D-	43			V
CMR04E470--D-	47			GJ
CMR04E510--D-	51			FGJ
CMR04E560--D-	56			↓
CMR04E620--D-	62			
CMR04E680--D-	68			
CMR04E750--D-	75			
CMR04E820--D-	82			
CMR04F910--D-	91			
CMR04F101--D-	100			
CMR04F111--D-	110			
CMR04F121--D-	120			
CMR04F131--D-	130			
CMR04F151--D-	150			
CMR04F161--D-	160			
CMR04F181--D-	180			
CMR04F201--D-	200			
CMR04F221--D-	220	↓		
CMR04F241--D-	240	500		
CMR04F271--C-	270	300		
CMR04F301--C-	300	300		
CMR04F331--A-	330	100		
CMR04F361--A-	360	100	↓	
CMR04F391--A-	390	100	OP	FGJ

1/ See footnote at end of table.

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TABLE II. Type designations - Continued.

STYLE CMR05				
Type designation 1/	Capacitance value	DC rated voltage	Operating temperature range	Capacitance tolerance available
	(pF)	(volts)		
CMR05C1R0D-D-	1.0	500	OP	D
CMR05C1R5D-D-	1.5	↓	↓	
CMR05C2R0D-D-	2.0			
CMR05C2R5D-D-	2.5			
CMR05C3R0D-D-	3.0			
CMR05C3R5D-D-	3.5			
CMR05C4R0D-D-	4.0			
CMR05C4R5D-D-	4.5			
CMR05C5R0D-D-	5.0			
CMR05C6R0D-D-	6.0			
CMR05C7R0D-D-	7.0			
CMR05C8R0D-D-	8.0			
CMR05C9R0D-D-	9.0			
CMR05C100D-D-	10			V
CMR05C110D-D-	11			D
CMR05C120J-D-	12			J
CMR05C150J-D-	15			
CMR05C180J-D-	18			
CMR05E200J-D-	20			
CMR05E220J-D-	22			V
CMR05E240J-D-	24			J
CMR05E270--D-	27			GJ
CMR05E300--D-	30			
CMR05E330--D-	33			
CMR05E360--D-	36			
CMR05E390--D-	39			
CMR05E430--D-	43			V
CMR05E470--D-	47			GJ
CMR05E510--D-	51			FGJ
CMR05E560--D-	56			
CMR05E620--D-	62			
CMR05E680--D-	68			
CMR05E750--D-	75			
CMR05E820--D-	82			
CMR05F910--D-	91			
CMR05F101--D-	100			
CMR05F111--D-	110			
CMR05F121--D-	120			
CMR05F131--D-	130			
CMR05F151--D-	150			
CMR05F161--D-	160			
CMR05F181--D-	180			
CMR05F201--D-	200			
CMR05F221--D-	220			
CMR05F241--D-	240			
CMR05F271--D-	270			
CMR05F301--D-	300			
CMR05F331--D-	330			
CMR05F361--D-	360	↓	↓	V
CMR05F391--D-	390	500	OP	FGJ

1/ See footnote at end of table.

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TABLE II. Type designations - Continued.

Type designation 1/	Capacitance value	DC rated voltage	Operating temperature range	Capacitance tolerance available
	(pF)	(volts)		
STYLE CMR06				
CMR06F431--D-	430	500	OP	FGJ
CMR06F471--D-	470			
CMR06F511--D-	510			
CMR06F561--D-	560			
CMR06F621--D-	620			
CMR06F681--D-	680			
CMR06F751--D-	750			
CMR06F821--D-	820			
CMR06F911--D-	910			
CMR06F102--D-	1,000			
CMR06F112--D-	1,100			
CMR06F122--D-	1,200			
CMR06F132--D-	1,300			
CMR06F152--D-	1,500			
CMR06F162--D-	1,600			
CMR06F182--D-	1,800			
CMR06F202--D-	2,000			
CMR06F222--D-	2,200			
CMR06F242--D-	2,400			
CMR06F272--D-	2,700			
CMR06F302--D-	3,000			
CMR06F332--D-	3,300			
CMR06F362--D-	3,600			
CMR06F392--D-	3,900			
CMR06F432--D-	4,300	↓	↓	↓
CMR06F472--D-	4,700	500	OP	FGJ
STYLE CMR07				
CMR07F512--D-	5,100	500	OP	FGJ
CMR07F562--D-	5,600			
CMR07F622--D-	6,200			
CMR07F682--D-	6,800			
CMR07F752--D-	7,500			
CMR07F822--D-	8,200			
CMR07F912--D-	9,100			
CMR07F103--D-	10,000			
CMR07F113--D-	11,000			
CMR07F123--D-	12,000			
CMR07F133--D-	13,000			
CMR07F153--D-	15,000			
CMR07F163--D-	16,000			
CMR07F183--D-	18,000	↓	↓	↓
CMR07F203--D-	20,000	500	OP	FGJ

1/ See footnote at end of table.

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TABLE II. Type designations - Continued.

STYLE CMR08				
Type designation 1/	Capacitance value	DC rated voltage	Operating temperature range	Capacitance tolerance available
	(pF)	(volts)		
CMR08F223--D-	22,000	500	OP	FGJ
CMR08F243--D-	24,000	↓	↓	↓
CMR08F273--D-	27,000			
CMR08F303--D-	30,000			
CMR08F333--D-	33,000			
CMR08F363--D-	36,000			
CMR08F393--D-	39,000			
CMR08F433--D-	43,000			
CMR08F473--D-	47,000	↓		
CMR08F513--D-	51,000	500		
CMR08F563--C-	56,000	300		
CMR08F623--C-	62,000	300		
CMR08F683--C-	68,000	300		
CMR08F753--A-	75,000	100		
CMR08F823--A-	82,000	100	↓	↓
CMR08F913--A-	91,000	100	OP	FGJ

1/ Complete type designation will include additional letter symbols to indicate the following, where applicable:
Capacitance tolerance: D (±.5 pF); F (±1%); G (±2%); or J (±5%).
Operating temperature range: O (-55° to +125°C) or P (-55° to +150°C).
Rated voltage: Y (50v); A (100v); C (300v); or D (500v).
Failure rate level: L (2.0%); M (1.0%); P (0.1%); R (0.01%); or S (0.001%).

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5910-1203)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget bureau No 22-R355

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-39001/5B CAPACITORS, FIXED, MICA DIELECTRIC, ESTABLISHED RELIABILITY, STYLES CMR03, CMR04, CMR06, CMR07, AND CMR08

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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